Algebra Monsoon 2026 Exercise : How to write

Proposition 1

Consider a diagram of mappings :

$$\begin{array}{ccc} A & \xrightarrow{\varphi_A} & A' \\ f \downarrow & & \\ B & \xrightarrow{\varphi_B} & B' \end{array}$$

Show that if φ_A and φ_B are bijections, then there is a unique mapping

$$f' : A' \longrightarrow B'$$

such that $f' \circ \varphi_A = \varphi_B \circ f$.

Proof

Existence of f':

Take $f' = \varphi_B \circ f \circ \varphi_A^{-1}$.

We then have $f' \circ \varphi_A = \varphi_B \circ f$.

Uniqueness of f':

Suppose we have

$$g \circ \varphi_A = \varphi_B \circ f$$

for some $g : A' \longrightarrow B'$.

Then

$$\underbrace{(g \circ \varphi_A) \circ \varphi_A^{-1}}_{= g} = \underbrace{\varphi_B \circ f \circ \varphi_A^{-1}}_{= f'}$$

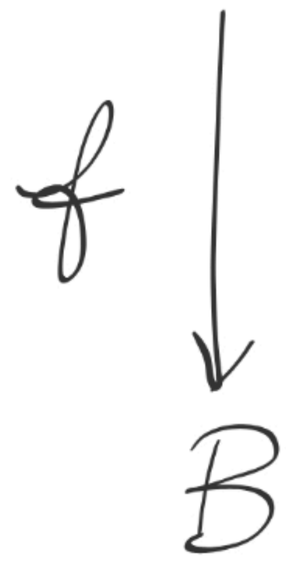
$\Rightarrow g = f'$ and f' is unique.

□

Simpler lemma which implies the above statement:

Proposition 2

Consider $A \xrightarrow{\varphi} A'$



If φ is <u>bijective</u>, there is a unique mapping

$$f' : A' \longrightarrow B$$

such that $f = f' \circ \varphi$.

<u>Proof</u>

<u>Existence of f'</u> :

Take $f' = f \circ \varphi^{-1}$.

We then have $f' \circ \varphi = (f \circ \varphi^{-1}) \circ \varphi$

$= f \circ (\varphi^{-1} \circ \varphi)$

$= f \circ Id_A$

$= f$.

<u>Uniqueness of f'</u> :

If we have $g \circ \varphi = f$

for some $g : A' \longrightarrow B$, then

$$g = g \circ Id_{A'} = g \circ (\varphi \circ \varphi^{-1})$$

$$= (g \circ \varphi) \circ \varphi^{-1}$$

$$= f \circ \varphi^{-1}$$

$$= f'.$$

□

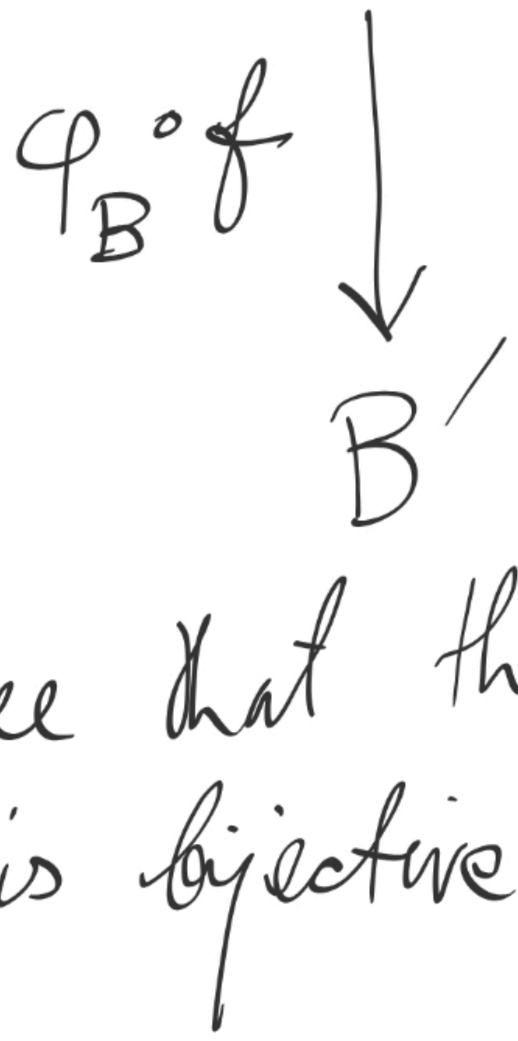
In fact, Proposition 1 is a corollary of Proposition 2 :

Given

$$\begin{array}{ccc} A & \xrightarrow{\varphi_A} & A' \\ f \downarrow & & \\ B & \xrightarrow{\varphi_B} & B' \end{array}$$

apply Proposition 2 to

$$A \xrightarrow{\varphi_A} A'$$

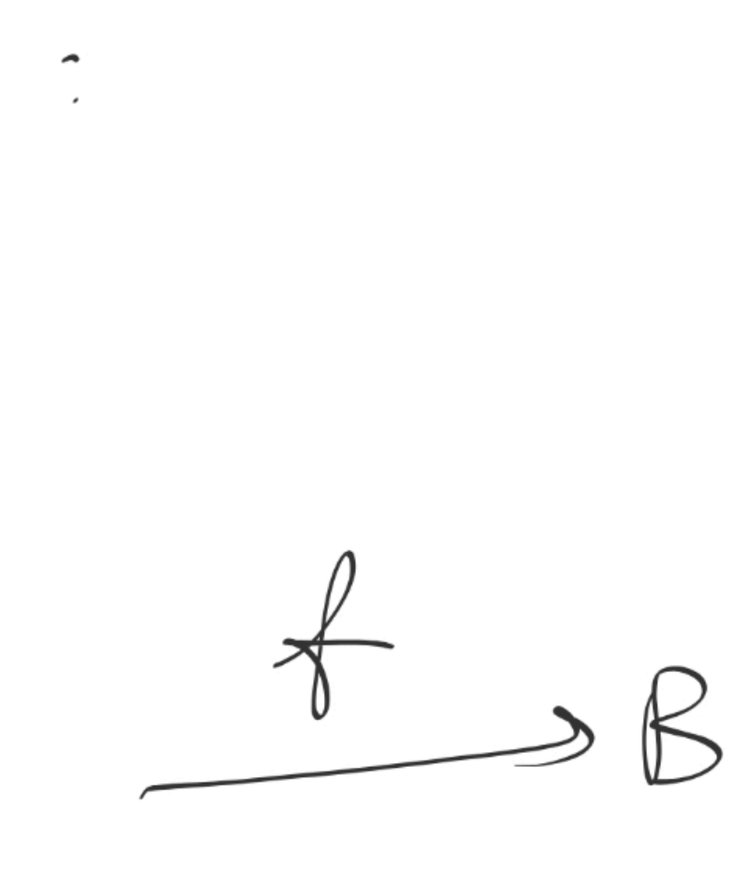


We see that the hypothesis
"φ_B is bijective" is not needed.

More generally :

Proposition 3

Given $A \xrightarrow{f} B$



,

- there exists a $\varphi : B \to C$
  such that $g = \varphi \circ f$
  $\iff \left(f(x) = f(y) \Longrightarrow g(x) = g(y) \right)$.
- Suppose that φ exists.
  φ is unique $\iff$ (f is surjective or
  C is a singleton).

Write your own proofs of these two statements.